

Information Circulation and External-Flow Suppression under Finite Capacity

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In complex systems, cyclic causal structures are closely associated with recurrent and oscillatory dynamics, yet an integrated dynamical description remains lacking. From an information-theoretic perspective, we formulate these structures as information circulation and show that, under the stated structural assumptions and fixed network support, a directed cycle is necessary for stationary or periodic entropic dynamics. We then analyze the merging of a pre-existing circulating flow and an external flow at a finite-capacity receiving node. Near the capacity boundary, increasing the circulating flow reduces the admissible upper bound on additional transfer entropy. This bound is formulated technically through the remaining conditional-entropy margin. Explicit receiver models illustrate both satisfaction and failure of the corresponding finite-range suppression condition. A stochastic two-bit network using a Fredkin-type gate attains the bound at the onset of interaction and exhibits a model-specific transient evolution. These results characterize how finite receiving capacity constrains external information transfer in systems with pre-existing circulation.

I. INTRODUCTION

Many physical, biological, and engineered systems process several competing signals through components with finite local entropy capacity. For instance, when a receiving component already participates in a cyclic information flow, a newly activated input must compete with the pre-existing circulation within the same finite state space. Although competition under finite capacity is intuitive and is closely related to optimization principles such as the information bottleneck [1, 2], it is not obvious how this competition is expressed

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dynamically at the level of directional information flow. This motivates a formulation in which cyclic causal relationships are described through information-dynamical quantities.

Cyclic causal relationships are widely observed in complex systems, including gene regulatory networks [3–7], thalamocortical brain circuits [8], Arctic climate feedbacks [9], and predator-prey ecosystems [10]. These structures can be interpreted as feedback loops that support directed information circulation and often exhibit oscillatory or recurrent behavior. However, a unified and statistically grounded description of such cyclic causal relationships remains lacking, particularly for analyzing their dynamical properties.

To address this issue, we adopt an information-theoretic framework, which provides a model-independent and universally applicable description of correlations and causality in probabilistic systems. Within this framework, cyclic causal relationships can be formulated as cycles in an information dynamical network.

Information dynamics has been extensively developed across multiple disciplines, such as information thermodynamics [11–15], quantum thermodynamics [16–18], neuroscience [19, 20], and bioinformatics [21]. These approaches describe entropy production in non-equilibrium systems and enable the identification of causal interactions in complex networks of neurons, genes, and proteins. In particular, mutual information and transfer entropy have been widely used to characterize correlations and directional information flow in many-body systems [13, 22–25].

Building on these developments, Ref. [26] established a general and exact framework for information dynamics in which individual entropy, mutual information, transfer entropy, and reversed transfer entropy [15, 27, 28] are treated as the dynamical variables governed by a set of evolution equations with control variables. In that framework, the zero-control limit was analyzed, revealing that nonzero control variables induce curvature in the evolution of these dynamical variables and reflect the statistical structure of the underlying system.

Despite its generality, this framework does not explicitly address interactions between information flows, which are necessary to describe the dynamical behavior and responses of

systems with cyclic structures.

In this paper, we extend the framework of Ref. [26] to describe cyclic causal relationships as cycles of transfer entropy, which encode directional causality [29]. Under locally deterministic dynamics and non-synergistic interactions, we demonstrate in Sec. III that a finite directed network of transfer entropy with a fixed support must contain a directed cycle in order to exhibit stationary or periodic information dynamics. This result identifies information circulation as a necessary structural feature of such dynamics under the present assumptions.

To further characterize these dynamics, Sec. IV analyzes the behavior of a single cycle within the information dynamical framework. While the stationary state of an isolated cycle was previously studied in Ref. [26], we introduce interactions between the cycle and an external information source to investigate how such interactions modify the dynamics.

The analysis reveals that the behavior of the system depends critically on how control variables are assigned at points where multiple information flows merge. Accordingly, Sec. V examines the information dynamics at such merging points. Under a specified set of assumptions, we show that increasing the pre-existing information flow reduces the admissible upper bound on additional incoming transfer entropy when the receiving node is near its finite-capacity boundary. Throughout this paper, external-flow suppression refers either to a reduction of this upper bound or, when explicitly demonstrated for a specific model, to a reduction of the realized external transfer entropy.

II. INFORMATION DYNAMICAL EQUATIONS

A. Definitions

We summarize the information-theoretic quantities in increasing order of structure, from pairwise measures to directional and multi-variable extensions.

Definition 1 *The **Shannon entropy** $H(X)$ and **conditional entropy** $H(Y|X)$ [30, 31]*

are defined as:

$$H(X) := - \sum_{x \in \mathcal{M}} P(X = x) \ln P(X = x) \quad (1)$$

$$H(Y|X) := H(X, Y) - H(X),$$

where X and Y are discrete random variables taking values in the state space $\mathcal{M}$.

Also, the entropy of a set of random variables $V = \{V_1, \dots, V_m\}$ is defined with the joint probability distribution:

$$H(V) := - \sum_{v_1, \dots, v_m \in \mathcal{M}} P(V_1 = v_1, \dots, V_m = v_m) \ln P(V_1 = v_1, \dots, V_m = v_m). \quad (2)$$

Definition 2 The **mutual information** between X and Y is defined as [31]:

$$I(X, Y) := H(X) + H(Y) - H(X, Y), \quad (3)$$

and the **conditional mutual information** is given by

$$I(X, Y|Z) := H(X|Z) + H(Y|Z) - H(X, Y|Z). \quad (4)$$

Definition 3 The **transfer entropy** [32, 33] from Y to X , which quantifies directional information flow, is defined as:

$$T_{Y \rightarrow X} := I(X', Y|X) \quad (5)$$

where X and X' are evaluated at time $t = t_0$ and $t = t_1$, respectively, with $t_1 > t_0$.

Throughout this paper, all primed variables in this paper are evaluated one time step after the indicated base time. For instance, X'' is evaluated at time $t = t_2$.

Definition 4 The **reversed transfer entropy** [15, 27, 28] (also referred to as backward transfer entropy [27]) is defined as,

$$T_{Y \rightarrow X}^\dagger := I(X, Y'|X'), \quad (6)$$

which corresponds to the transfer entropy evaluated under time-reversed indexing. All quantities defined above are non-negative.

Definition 5 The **co-information** (also called interaction information or multiple mutual information) [34–41] captures multi-correlations, such as synergy and redundancy, and is defined as:

$$I(V) := \sum_{i=1}^{|V|} (-1)^{i+1} \sum_{U \subset V, |U|=i} H(U). \quad (7)$$

Similarly, the **conditional co-information** is,

$$I(V|W) := \sum_{i=1}^{|V|} (-1)^{i+1} \sum_{U \subset V, |U|=i} H(U|W). \quad (8)$$

Definition 6 The **multiple transfer entropy** and **multiple reversed transfer entropy** [26] are defined as,

$$\begin{aligned} T_{V \rightarrow U} &:= I(V \cup U' | U) \\ T_{V \rightarrow U}^\dagger &:= I(V' \cup U | U'). \end{aligned} \quad (9)$$

B. The General Framework of Information Dynamics

For a general multi-variable system with N components, we formulate the dynamical equations[26] governing the temporal evolution of the information-theoretic quantities defined in Sec. II A. In this framework, the entropy, mutual information, transfer entropy, and reversed transfer entropy are treated as state variables whose temporal changes are described by a set of coupled evolution equations.

Let $\Gamma_1, \dots, \Gamma_N$ denote the random variables associated with N components and $\Delta H(\Gamma_i)$, $\Delta I(\Gamma_i, \Gamma_j)$, $\Delta T_{\Gamma_j \rightarrow \Gamma_i}$, and $\Delta T_{\Gamma_j \rightarrow \Gamma_i}^\dagger$ denote the temporal increments of the corresponding quantities over a discrete time step.

$$\left\{ \begin{aligned} \Delta H(\Gamma_i) &= \sum_{k \in K_i} [T_{k \rightarrow \Gamma_i} - T_{k \rightarrow \Gamma_i}^\dagger] - E_{K_i \rightarrow \Gamma_i} + \alpha_1^{K_i \rightarrow \Gamma_i} \\ \Delta I(\Gamma_i, \Gamma_j) &= [T_{\Gamma_j \rightarrow \Gamma_i} - T_{\Gamma_j \rightarrow \Gamma_i}^\dagger] + [T_{\Gamma_i \rightarrow \Gamma_j} - T_{\Gamma_i \rightarrow \Gamma_j}^\dagger] + \alpha_2^{\Gamma_i, \Gamma_j} \\ \Delta T_{\Gamma_j \rightarrow \Gamma_i} &= \sum_{k \in K_j - \{\Gamma_i\}} [T_{k \rightarrow \Gamma_j} - T_{k \rightarrow \Gamma_j}^\dagger] - [T_{\Gamma_j \rightarrow \Gamma_i} - T_{\Gamma_j \rightarrow \Gamma_i}^\dagger] - E_{K_j \rightarrow \Gamma_j} + \alpha_1^{K_j \rightarrow \Gamma_j} - \alpha_2^{\Gamma_i, \Gamma_j} - \alpha_3^{\Gamma_j \rightarrow \Gamma_i} \\ \Delta T_{\Gamma_j \rightarrow \Gamma_i}^\dagger &= \sum_{k \in K_j - \{\Gamma_i\}} [T_{k \rightarrow \Gamma_j} - T_{k \rightarrow \Gamma_j}^\dagger] - [T_{\Gamma_j \rightarrow \Gamma_i} - T_{\Gamma_j \rightarrow \Gamma_i}^\dagger] - E_{K_j \rightarrow \Gamma_j} + \alpha_1^{K_j \rightarrow \Gamma_j} - \alpha_2^{\Gamma_i, \Gamma_j} \\ &\quad + \alpha_4^{\Gamma_j} - \alpha_5^{\Gamma_i, \Gamma_j} - \alpha_6^{\Gamma_j \rightarrow \Gamma_i}, \end{aligned} \right. \quad (10)$$

with,

$$\begin{aligned}
\alpha_1^{K_i \rightarrow \Gamma_i} &:= H(\Gamma'_i | \Gamma_i, K_i) - H(\Gamma_i | \Gamma'_i, K'_i) \\
\alpha_2^{\Gamma_i, \Gamma_j} &:= I(\Gamma'_i, \Gamma'_j | \Gamma_i, \Gamma_j) - I(\Gamma_i, \Gamma_j | \Gamma'_i, \Gamma'_j) \\
\alpha_3^{\Gamma_j \rightarrow \Gamma_i} &:= \Delta H(\Gamma_j | \Gamma_i, \Gamma'_i) \\
\alpha_4^{\Gamma_j} &:= \Delta^2 H(\Gamma_j) = \Delta(\Delta H(\Gamma_j)) = H(\Gamma''_j) - 2H(\Gamma'_j) + H(\Gamma_j) \\
\alpha_5^{\Gamma_i, \Gamma_j} &:= \Delta^2 I(\Gamma_i, \Gamma_j) = I(\Gamma''_i, \Gamma''_j) - 2I(\Gamma'_i, \Gamma'_j) + I(\Gamma_i, \Gamma_j) \\
\alpha_6^{\Gamma_j \rightarrow \Gamma_i} &:= \Delta H(\Gamma'_j | \Gamma_i, \Gamma'_i).
\end{aligned} \tag{11}$$

Here, K_i is a set of causal parents of component i and $K'_i := \{k' | k \in K_i\}$.

The term $E_{K_i \rightarrow \Gamma_i}$ is responsible for the coupling between first-order(pairwise) correlations and high-order correlations: for $V, U, W \subset \{\Gamma_1, \dots, \Gamma_N\}$,

$$\left\{ \begin{aligned}
E_{K_i \rightarrow \Gamma_i} &:= \sum_{u=2}^{|K_i|} (-1)^u \sum_{V \subset K_i, |V|=u} [T_{V \rightarrow \Gamma_i} - T_{V \rightarrow \Gamma_i}^\dagger] \\
\Delta I(V) &= \sum_{(U, W) \in D_V} [T_{U \rightarrow W} - T_{U \rightarrow W}^\dagger] + \alpha_2^V \\
\Delta T_{U \rightarrow W} &= (-1)^{|W|} \Delta I(U \cup W) + \sum_{u=1}^{|W|} (-1)^{|W|-u} \sum_{S \subset W, |S|=u} (\Delta I(U \cup (W \setminus S))) \\
&\quad + \sum_{s=1}^u (-1)^s \sum_{Z \subset S, |Z|=s} \alpha_3^{U \cup (W \setminus S) \rightarrow Z}) \\
\Delta T_{U \rightarrow W}^\dagger &= (-1)^{|W|} \Delta I(U \cup W) + (-1)^{|W|} \alpha_5^{U \cup W} + \sum_{u=1}^{|W|} (-1)^{|W|-u} \sum_{S \subset W, |S|=u} (\Delta I(U \cup (W \setminus S))) \\
&\quad + \alpha_5^{U \cup (W \setminus S)} + \sum_{s=1}^u (-1)^s \sum_{Z \subset S, |Z|=s} \alpha_6^{U \cup (W \setminus S) \rightarrow Z}),
\end{aligned} \right. \tag{12}$$

with,

$$\begin{aligned}
\alpha_2^V &:= I(V' | V) - I(V | V') \\
\alpha_3^{X \rightarrow Z} &:= \Delta I(X | Z \cup Z') \\
\alpha_5^V &:= \Delta^2 I(V) \\
\alpha_6^{X \rightarrow Z} &:= \Delta I(X' | Z \cup Z').
\end{aligned} \tag{13}$$

Each equation is exact [26] and consists of an intrinsic contribution, determined by the probabilistic structure of the system, and a control contribution, parameterized by

coefficients α , which encode deviations from the zero-control limit. In the zero-control limit, these equations reduce to the intrinsic dynamics analyzed in Ref. [26]. Furthermore, the control variables α encode how interaction structures and constraints modify the intrinsic information dynamics. Their values are determined by mechanisms such as blocked channels and finite-capacity effects at merging points.

Ref. [26] demonstrated that the non-trivial circulation near the zero-control limit arises from the restriction to a one-way interaction channel between component A and B :

$$\begin{cases} [T_{B \rightarrow A} = 0 \text{ and } T_{A \rightarrow B}^\dagger = 0] \\ \text{or } [T_{A \rightarrow B} = 0 \text{ and } T_{B \rightarrow A}^\dagger = 0]. \end{cases} \quad \text{for all time} \quad (14)$$

Eq. (14) is referred to as the blocking-flows condition.

Furthermore, this paper postulates Eq. (14) along with the stronger condition that $T_{B \rightarrow A} \neq 0$ if and only if $T_{A \rightarrow B}^\dagger \neq 0$, denoting the strong blocking flows condition. This concept is readily established in statistical modeling, as illustrated by the example in Ref. [26].

C. Structural Assumption : Deterministic Reversible Dynamics

The analysis of Sec. III assumes the deterministic reversible dynamics: $\exists U = (U_1, U_2, \dots, U_N)$ and U^{-1} such that,

$$\begin{aligned} \Gamma'_i &= U_i(\Gamma_1, \dots, \Gamma_N, t_0) \\ \Gamma_i &= (U^{-1})_i(\Gamma'_1, \dots, \Gamma'_N, t_1), \end{aligned} \quad (15)$$

where $i \in \{1, 2, \dots, N\}$ and $t_1 > t_0$. Then, with $K_i^{total} := \{\Gamma_1, \dots, \Gamma_N\} - \{\Gamma_i\}$, the following uncertainty vanishes:

$$\begin{aligned} H(\Gamma'_i | \Gamma_i, K_i^{total}) &= 0 \\ H(\Gamma_i | \Gamma'_i, K_i^{total}) &= 0. \end{aligned} \quad (16)$$

Hence, the control parameter $\alpha_1^{K_i^{total} \rightarrow \Gamma_i}$ defined in the equation (11) becomes zero in this case:

$$\alpha_1^{K_i^{total} \rightarrow \Gamma_i} := H(\Gamma'_i | \Gamma_i, K_i^{total}) - H(\Gamma_i | \Gamma'_i, K_i^{total}) = 0. \quad (17)$$

Furthermore, the set K_i can be effectively shrunk to the set of the causal parents: $K_i := \{k_1, \dots, k_m\} \subset \{\Gamma_1, \dots, \Gamma_N\}$, such that,

$$\begin{aligned} \Gamma'_i &= U_i(\Gamma_i, k_1, \dots, k_m) \\ \Gamma_i &= (U^{-1})_i(\Gamma'_i, k'_1, \dots, k'_m). \end{aligned} \quad (18)$$

Under the assumption of deterministic reversible dynamics, such a set exists, since the set K_i^{total} satisfies Eq. (18). Hence, by subtracting one by one from K_i^{total} while preserving Eq. (18), the smaller set of the causal parents is obtained. Such a set need not be unique. For each component i , we choose one set K_i satisfying both relations in Eq. (18), and this choice is held fixed throughout Section III.

As a result,

$$\alpha_1^{K_i \rightarrow \Gamma_i} := H(\Gamma'_i | \Gamma_i, K_i) - H(\Gamma_i | \Gamma'_i, K_i) = 0. \quad (19)$$

D. Structural Assumption : Absence of Synergistic Interactions

Let the neighbor set of a component v be $K = \{k_1, \dots, k_m\}$, and define an arbitrary but fixed tuple $\mathbf{K} := (k)_{k \in K}$. Then, the transfer entropy and reversed transfer entropy,

$$\begin{aligned} T_{\mathbf{K} \rightarrow v} &:= I(\mathbf{K}, v' | v) \\ T_{\mathbf{K} \rightarrow v}^\dagger &:= I(\mathbf{K}', v | v'), \end{aligned} \quad (20)$$

are distinct from the multiple transfer entropy $T_{K \rightarrow \{v\}}$ and multiple reversed transfer entropy $T_{K \rightarrow \{v\}}^\dagger$ defined in Section II A.

Decomposing $T_{\mathbf{K} \rightarrow v}$ and $T_{\mathbf{K} \rightarrow v}^\dagger$ yields:

$$\begin{aligned} T_{\mathbf{K} \rightarrow v} &= T_{k \rightarrow v} + I(\mathbf{K}_{\setminus k}, v' | v, k) \\ T_{\mathbf{K} \rightarrow v}^\dagger &= T_{k \rightarrow v}^\dagger + I(\mathbf{K}'_{\setminus k}, v | v', k'), \end{aligned} \quad (21)$$

for $k \in K$.

Since the right hand terms are non-negative, $T_{k \rightarrow v} = 0$ for all $k \in K$ if $T_{\mathbf{K} \rightarrow v} = 0$, and $T_{k \rightarrow v}^\dagger = 0$ for all $k \in K$ if $T_{\mathbf{K} \rightarrow v}^\dagger = 0$. Nevertheless, the converse is not generally true, as the

counterexample is presented in Appendix A. Restricting to non-synergistic interaction, the analysis of Sec. III assumes that the converse holds, i.e., if $T_{k \rightarrow v} = 0$ for all $k \in K$, then $T_{\mathbf{K} \rightarrow v} = 0$, and if $T_{k \rightarrow v}^\dagger = 0$ for all $k \in K$, $T_{\mathbf{K} \rightarrow v}^\dagger = 0$. This assumption excludes the pure synergistic effect, such as the XOR operation (Appendix A), while including, for example, the ternary additive receiver (Sec. VD 1).

III. GENERAL INFORMATION DYNAMICS OF A FINITE SYSTEM

We consider an N -component system satisfying the strong blocking flows condition and the assumption of deterministic reversible dynamics (Sec. II C). The system can be represented as a directed finite network, where directed edges correspond to nonzero pairwise transfer entropy (or reversed transfer entropy) from fixed causal parents satisfying Eq. (18). Under the strong blocking flows condition, the pairwise transfer entropy and reversed transfer entropy networks have mutually reversed edge sets. To analyze the role of network structure, we distinguish between acyclic networks and networks containing at least one cycle. We recall the following definitions from graph theory [42, 43].

Definition 7 A **directed graph** is a pair $G = (V, \vec{E})$, where V is a finite set of vertices and $\vec{E} \subset \{(x, y) \in V \times V | x \neq y\}$ is a set of directed edges. Also, the reversed edge set $\overleftarrow{E} := \{(y, x) | (x, y) \in \vec{E}\}$ is defined. Then, the **in-degree** of a vertex $v \in V$ is $d^-(v) = |\{x \in V | (x, v) \in \vec{E}\}|$, while the **out-degree** of a vertex $v \in V$ is $d^+(v) = |\{x \in V | (v, x) \in \vec{E}\}|$.

Throughout this section, a directed graph $G = (V, \vec{E})$ is assumed to have no isolated vertices; that is, every vertex $v \in V$ satisfies $d^-(v) + d^+(v) > 0$.

Definition 8 A **source** in a directed graph $G = (V, \vec{E})$ is the vertex $v \in V$ with $d^-(v) = 0$ and $d^+(v) \neq 0$, and a **sink** in a directed graph $G = (V, \vec{E})$ is the vertex $v \in V$ with $d^+(v) = 0$ and $d^-(v) \neq 0$.

Definition 9 A **cycle** C in a directed graph $(V, \vec{E})$ is a sequence of distinct vertices $v_1, \dots, v_n$, with $n \geq 2$, such that $(v_i, v_{i+1}) \in \vec{E}$ for $i = 1, \dots, n-1$, and $(v_n, v_1) \in \vec{E}$.

Definition 10 A directed graph is **acyclic** if it contains no cycle.

For each component v , let $K_v \subset V \setminus \{v\}$ denote the set of causal parents of v , namely the components whose states enter the local update rule of v as Eq. (18). Then, a directed graph of pairwise transfer entropy is defined as a directed graph $G_T = (V, \vec{E})$ with $\vec{E} = \{(x, y) \in V \times V | x \neq y, x \in K_y, T_{x \rightarrow y} > 0\}$.

Propositions 1 and 2 show that acyclic information networks necessarily exhibit monotonic entropy evolution, preventing stationary or periodic entropic dynamics under the present assumptions.

Proposition 1 An acyclic graph must include at least one source and one sink.

(proof) From Proposition (3.19) in the chapter "Directed Acyclic Graphs and Topological Ordering" of Ref. [43], a finite acyclic graph $G = (V, \vec{E})$ must include at least one source. Since the directed graph with reversed edges $G^{rev} = (V, \overleftarrow{E})$ is also acyclic, it has a source w that is the sink in the original graph $G = (V, \vec{E})$. ■

While Proposition 1 is a classical result of graph theory, Proposition 2 focuses on the information dynamical consequences within the framework of II B, implying that, in acyclic information networks, individual entropy cannot be stationary or periodic.

Proposition 2 Let $G_T = (V, \vec{E})$ be a directed network of pairwise transfer entropy with fixed support. Under the standing assumptions of this section and the non-synergistic assumption of Sec. II D, if G_T is acyclic, then the individual entropy of a sink evolves monotonically in time.

(proof) From Proposition 1, G_T has a sink $v \in V$: $\exists k \in K_v$ such that $T_{k \rightarrow v} > 0$, where K_v is the fixed causal-parent set of v . Then, the dynamics of v 's individual entropy (Eq. (10)) is,

$$\begin{aligned} \Delta H(v) &= \sum_{k \in K_v} [T_{k \rightarrow v} - T_{k \rightarrow v}^\dagger] - E_{K_v \rightarrow v} + \alpha_1^{K_v \rightarrow v} \\ &= \sum_{k \in K_v} [T_{k \rightarrow v} - T_{k \rightarrow v}^\dagger] - E_{K_v \rightarrow v} \\ &= T_{\mathbf{K}_v \rightarrow v} - T_{\mathbf{K}_v \rightarrow v}^\dagger, \end{aligned} \tag{22}$$

where $\mathbf{K}_v := (k)_{k \in K_v}$.

Here, $\alpha_1^{K_v \rightarrow v} = 0$ due to the assumption of deterministic and reversible dynamics.

Since v is a sink, $T_{v \rightarrow k} = 0$ for every $k \in K_v$. By the strong blocking flows condition, $T_{k \rightarrow v}^\dagger > 0$ would imply $T_{v \rightarrow k} > 0$, contradicting the sink property. Hence, $T_{k \rightarrow v}^\dagger = 0$ for every $k \in K_v$. The reversed non-synergy assumption of Section II D requires $T_{\mathbf{K}_v \rightarrow v}^\dagger = 0$. In addition, by definition, $T_{\mathbf{K}_v \rightarrow v} := I(\mathbf{K}_v, v' | v) \geq 0$. If $T_{\mathbf{K}_v \rightarrow v} = 0$, then $T_{k \rightarrow v} = 0$ for all $k \in K_v$ by the argument in Section II D, which contradicts the fact that v is the sink.

Therefore,

$$\Delta H(v) = T_{\mathbf{K}_v \rightarrow v} > 0. \tag{23}$$

■

Since $\Delta H(v) > 0$ at every step, $H(v)$ is strictly increasing and therefore cannot be constant or periodic over any interval on which the support and the standing assumptions remain valid. If bounded, $H(v)$ may nevertheless converge monotonically to a limiting value. Consequently, one can conclude that the information-dynamical entropic state of a whole system, $\{H(u)\}_{u \in V}$, cannot be stationary or periodic in this acyclic case, as the system has a node with such monotonic behavior. Therefore, we restrict our analysis to finite information networks with cycles to investigate other kinds of dynamical behaviors.

IV. INFORMATION DYNAMICS OF A FINITE SYSTEM WITH A CYCLE

The information dynamics of finite systems containing a cycle is analyzed within the framework of Sec. II, focusing on the behavior of a single cycle and its interaction with

an external information source. Prior to that, the dynamics of an isolated cycle is considered.

A. The Stationary State of a Single Cycle

Let an N -component system $\Gamma = \{\Gamma_0, \Gamma_1, \dots, \Gamma_{N-1}\}$ form a one-dimensional system with periodic boundary condition. In the index notation, $\text{mod } N$ is assumed in this section. By imposing the blocking flows condition and near the zero-control limit, Ref. [26] obtains,

$$\begin{aligned}
 \Delta H(\Gamma_i) &= T_{\Gamma_{i-1} \rightarrow \Gamma_i} - T_{\Gamma_{i+1} \rightarrow \Gamma_i}^\dagger \\
 \Delta I(\Gamma_i, \Gamma_{i+1}) &= T_{\Gamma_i \rightarrow \Gamma_{i+1}} - T_{\Gamma_{i+1} \rightarrow \Gamma_i}^\dagger \\
 \Delta T_{\Gamma_i \rightarrow \Gamma_{i+1}} &= T_{\Gamma_{i-1} \rightarrow \Gamma_i} - T_{\Gamma_i \rightarrow \Gamma_{i+1}} \\
 \Delta T_{\Gamma_i \rightarrow \Gamma_{i+1}}^\dagger &= 0 \\
 \Delta T_{\Gamma_i \rightarrow \Gamma_{i-1}} &= 0 \\
 \Delta T_{\Gamma_{i+1} \rightarrow \Gamma_i}^\dagger &= T_{\Gamma_i \rightarrow \Gamma_{i-1}}^\dagger - T_{\Gamma_{i+1} \rightarrow \Gamma_i}^\dagger.
 \end{aligned} \tag{24}$$

These equations describe the circulation of transfer entropy and reversed transfer entropy along the cycle. A stationary state is achieved when the transfer entropy and reversed transfer entropy are uniformly distributed along the cycle, satisfying $T_{\Gamma_{i-1} \rightarrow \Gamma_i} = T_{\Gamma_i \rightarrow \Gamma_{i+1}}$, $T_{\Gamma_i \rightarrow \Gamma_{i-1}}^\dagger = T_{\Gamma_{i+1} \rightarrow \Gamma_i}^\dagger$, and $T_{\Gamma_{i-1} \rightarrow \Gamma_i} = T_{\Gamma_{i+1} \rightarrow \Gamma_i}^\dagger$. This corresponds to a constant circulation of information along the cycle. As shown in Ref. [44], small deviations from this stationary state remain bounded in the examples analyzed there.

B. The Transient Behaviors of the Single Cycle and a Random Source

Consider the interaction between a single cycle and an external information source. Let an external component Ext begin to interact with a component p in the cycle at the time $t = t_1$ as shown in Fig. 1, while no interaction is present at earlier times. Therefore, the stationary state of the previous section is assumed for the time $t = t_0 < t_1$. Then, the resulting dynamics is described by Proposition 3.

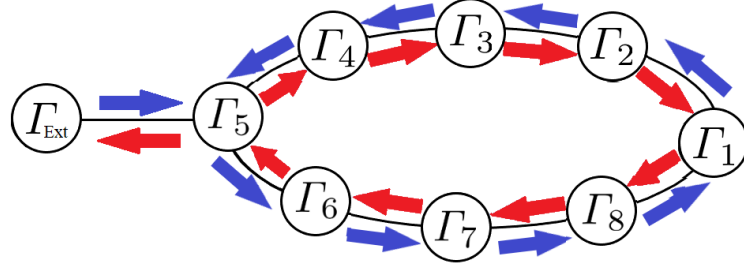


FIG. 1: Schematic illustrations of information dynamics on a cycle interacting with a random source (*Ext*). Blue arrows represent transfer entropy, and red arrows represent reversed transfer entropy.

Proposition 3 *The resulting dynamical equations for the aforementioned case are given by:*

$$\begin{aligned}
\Delta H(\Gamma_i) &= T_{\Gamma_{i-1} \rightarrow \Gamma_i} - T_{\Gamma_{i+1} \rightarrow \Gamma_i}^\dagger \quad (i \neq p) \\
\Delta H(\Gamma_p) &= T_{\Gamma_{p-1} \rightarrow \Gamma_p} + T_{\Gamma_{Ext} \rightarrow \Gamma_p} - T_{\Gamma_{p+1} \rightarrow \Gamma_p}^\dagger - E_{\{\Gamma_{p-1}, \Gamma_{p+1}, \Gamma_{Ext}\} \rightarrow \Gamma_p} \\
\Delta I(\Gamma_i, \Gamma_{i+1}) &= T_{\Gamma_i \rightarrow \Gamma_{i+1}} - T_{\Gamma_{i+1} \rightarrow \Gamma_i}^\dagger \\
\Delta I(\Gamma_p, \Gamma_{Ext}) &= T_{\Gamma_{Ext} \rightarrow \Gamma_p} - T_{\Gamma_p \rightarrow \Gamma_{Ext}}^\dagger \\
\\
\Delta T_{\Gamma_i \rightarrow \Gamma_{i+1}} &= T_{\Gamma_{i-1} \rightarrow \Gamma_i} - T_{\Gamma_i \rightarrow \Gamma_{i+1}} \quad (i \neq p-1, p) \\
\Delta T_{\Gamma_i \rightarrow \Gamma_{i+1}}^\dagger &= 0 \\
\Delta T_{\Gamma_i \rightarrow \Gamma_{i-1}} &= 0 \\
\Delta T_{\Gamma_i \rightarrow \Gamma_{i-1}}^\dagger &= T_{\Gamma_{i-1} \rightarrow \Gamma_{i-2}}^\dagger - T_{\Gamma_i \rightarrow \Gamma_{i-1}}^\dagger \quad (i \neq p, p+1) \\
\\
\Delta T_{\Gamma_{p-1} \rightarrow \Gamma_p} &= T_{\Gamma_{p-2} \rightarrow \Gamma_{p-1}} - T_{\Gamma_{p-1} \rightarrow \Gamma_p} - \alpha_3^{\Gamma_{p-1} \rightarrow \Gamma_p} \\
\Delta T_{\Gamma_p \rightarrow \Gamma_{p-1}}^\dagger &= T_{\Gamma_{p-1} \rightarrow \Gamma_{p-2}}^\dagger - T_{\Gamma_p \rightarrow \Gamma_{p-1}}^\dagger - E'_{\{\Gamma_{p-1}, \Gamma_{p+1}, \Gamma_{Ext}\} \rightarrow \Gamma_p} \\
\Delta T_{\Gamma_p \rightarrow \Gamma_{p+1}} &= T_{\Gamma_{p-1} \rightarrow \Gamma_p} + T_{\Gamma_{Ext} \rightarrow \Gamma_p} - T_{\Gamma_p \rightarrow \Gamma_{p+1}} - E_{\{\Gamma_{p-1}, \Gamma_{p+1}, \Gamma_{Ext}\} \rightarrow \Gamma_p} \\
\Delta T_{\Gamma_{p+1} \rightarrow \Gamma_p}^\dagger &= \alpha_1^{\{\Gamma_p\} \rightarrow \Gamma_{Ext}} - \alpha_3^{\Gamma_{Ext} \rightarrow \Gamma_p} + T_{\Gamma_p \rightarrow \Gamma_{p-1}}^\dagger - T_{\Gamma_{p+1} \rightarrow \Gamma_p}^\dagger \\
&\quad - E'_{\{\Gamma_{p-1}, \Gamma_{p+1}, \Gamma_{Ext}\} \rightarrow \Gamma_p} - \alpha_6^{\Gamma_p \rightarrow \Gamma_{p-1}} \\
\\
\Delta T_{\Gamma_{Ext} \rightarrow \Gamma_p} &= \alpha_1^{\{\Gamma_p\} \rightarrow \Gamma_{Ext}} - T_{\Gamma_{Ext} \rightarrow \Gamma_p} - \alpha_3^{\Gamma_{Ext} \rightarrow \Gamma_p} \\
\Delta T_{\Gamma_p \rightarrow \Gamma_{Ext}}^\dagger &= -\alpha_{3456}^{\Gamma_p \rightarrow \Gamma_{Ext}},
\end{aligned} \tag{25}$$

under the zero-control limit,

$$\begin{aligned}
\alpha_1^{\{\Gamma_{i-1}, \Gamma_{i+1}\} \rightarrow \Gamma_i} &= \alpha_2^{\Gamma_i, \Gamma_{i+1}} = 0 \quad (i \neq p) \\
\alpha_1'^{\{\Gamma_{i-1}, \Gamma_{i+1}\} \rightarrow \Gamma_i} &= \alpha_2'^{\Gamma_i, \Gamma_{i+1}} = 0 \quad (i \neq p) \\
\alpha_1^{\{\Gamma_{p-1}, \Gamma_{p+1}, \Gamma_{Ext}\} \rightarrow \Gamma_p} &= \alpha_1'^{\{\Gamma_{p-1}, \Gamma_{p+1}, \Gamma_{Ext}\} \rightarrow \Gamma_p} = \alpha_2^{\Gamma_i, \Gamma_{i+1}} = \alpha_2'^{\Gamma_i, \Gamma_{i+1}} = 0 \\
\alpha_3^{\Gamma_i \rightarrow \Gamma_{i+1}} &= 0 \quad (i \neq p-1) \\
\alpha_6^{\Gamma_{i-1} \rightarrow \Gamma_{i-2}} &= 0 \quad (i \neq p+1) \\
\alpha_3^{\{\Gamma_{i-1}, \Gamma_{i+1}\} \rightarrow \Gamma_i} &= \alpha_5^{\{\Gamma_{i-1}, \Gamma_{i+1}\}} = \alpha_5^{\{\Gamma_{i-1}, \Gamma_i, \Gamma_{i+1}\}} = \alpha_6^{\{\Gamma_{i-1}, \Gamma_{i+1}\} \rightarrow \Gamma_i} = 0 \quad (i \neq p).
\end{aligned} \tag{26}$$

All zero-control conditions in Eq. (26) are imposed at every time step over the interval

considered, and the initial conditions $E_{\{\Gamma_{i-1}, \Gamma_{i+1}\} \rightarrow \Gamma_i} = 0$ for all $i \neq p$ are assumed.

(proof) Appendix B.

These equations show that the external source modifies the information flow locally at component p , where the incoming flow $T_{\Gamma_{Ext} \rightarrow \Gamma_p}$ merges with the circulating flow $T_{\Gamma_{p-1} \rightarrow \Gamma_p}$. The corresponding control variables determine how the newly activated interaction is incorporated into the pre-existing circulation.

Specifically, the terms, $\alpha_3^{\Gamma_{p-1} \rightarrow \Gamma_p}$, $\alpha_6^{\Gamma_p \rightarrow \Gamma_{p-1}}$, $\alpha_3^{\Gamma_{Ext} \rightarrow \Gamma_p}$, and $\alpha_6^{\Gamma_p \rightarrow \Gamma_{Ext}}$, are expected to become nonzero during the transient period after $t = t_1$, reflecting the local reorganization of information flow at the merging component p . Since the transfer entropy $T_{\Gamma_{Ext} \rightarrow \Gamma_p} = 0$ for $t = t_0 < t_1$, the control variable $\alpha_3^{\Gamma_{Ext} \rightarrow \Gamma_p}$ determines the emergence of the nonzero transfer entropy $T_{\Gamma_{Ext} \rightarrow \Gamma_p}$ at later times. This interaction can therefore be interpreted as the merging of two information flows, $T_{\Gamma_{Ext} \rightarrow \Gamma_p}$ and $T_{\Gamma_{p-1} \rightarrow \Gamma_p}$, at the component p . The control variables $\alpha_3^{\Gamma_{Ext} \rightarrow \Gamma_p}$ and $\alpha_3^{\Gamma_{p-1} \rightarrow \Gamma_p}$ govern the dynamics of this merging process.

It should be noted that Eq. (25) applies when a stochastic process satisfies the zero-control limit and initial condition specified in Proposition 3. Beyond its practical utility, however, one must question whether an arbitrary trajectory of Eq. (25) corresponds to a realizable sequence of joint probability distributions, as it is derived from a purely dynamical perspective. In Section V B, we demonstrate that cases exist where such realizability fails to hold. Nevertheless, this paper regards Eq. (25) as an applicable framework for statistically realizable systems, reserving the characterization of the admissible range of control parameters for future investigation.

V. THE MERGING OF TWO INFORMATION FLOWS

When a cycle of information flow interacts with an external information source, the manner in which two information flows merge at a single component determines the resulting dynamics. The merging of two information flows is analyzed within the framework of Sec. II B.

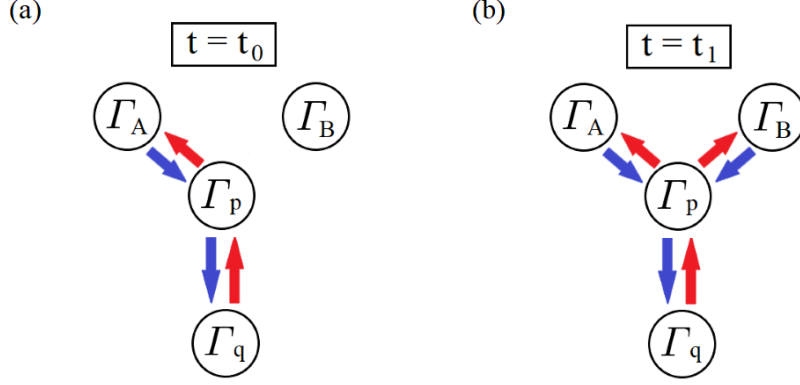


FIG. 2: Schematic illustration of the merging of two information flows at time **(a)** $t = t_0$ and **(b)** $t = t_1$. Blue arrows denote transfer entropy, and red arrows denote reversed transfer entropy.

A. Problem Setting

Let A, B, P and Q be random variables defined over an ensemble, representing the states of four components. We consider a configuration in which the components represented by A and B interact with the component represented by P , while P also interacts with the component represented by Q (Fig. 2). At time $t = t_0$, the components A, P , and Q interact with each other, with the nonzero stationary transfer entropy $T_{A \rightarrow P} = T_{P \rightarrow Q} = \mu_A \neq 0$ and reversed transfer entropy $T_{P \rightarrow A}^\dagger = T_{Q \rightarrow P}^\dagger = \mu_A$, while the component B is not interacting with the component P , implying $T_{B \rightarrow P} = 0$ and $T_{P \rightarrow B}^\dagger = 0$. Then, the components B and P start to interact at the time $t = t_1$. The quantities $T_{B' \rightarrow P'}$ and $T_{P' \rightarrow B'}^\dagger$ are determined from the dynamical equations (Eq. (10)).

$$\begin{aligned}
 T_{B' \rightarrow P'} &= \Delta T_{B \rightarrow P} = \sum_{k \in K_B \setminus \{P\}} [T_{k \rightarrow B} - T_{k \rightarrow B}^\dagger] \\
 &\quad - E_{K_B \rightarrow B} + \alpha_1^{K_B \rightarrow B} - \alpha_2^{B, P} - \alpha_3^{B \rightarrow P}, \\
 T_{P' \rightarrow B'}^\dagger &= \Delta T_{P \rightarrow B}^\dagger \\
 &= [T_{A \rightarrow P} - T_{Q \rightarrow P}^\dagger] - E_{\{A, B, Q\} \rightarrow P} \\
 &\quad + \alpha_1^{\{A, B, Q\} \rightarrow P} - \alpha_2^{B, P} + \alpha_4^P - \alpha_5^{B, P} - \alpha_6^{P \rightarrow B}.
 \end{aligned} \tag{27}$$

The conditions $T_{B \rightarrow P} = T_{P \rightarrow B}^\dagger = 0$ and the stationary conditions $\Delta I(B, P) = 0$ imply that $\alpha_2^{B,P} = 0$. Moreover, $E_{\{A,B,Q\} \rightarrow P} = E_{\{A,Q\} \rightarrow P} = 0$, as the component B does not interact with component P at time $t = t_0$.

$$\begin{aligned}
T_{B' \rightarrow P'} &= \sum_{k \in K_B \setminus \{P\}} [T_{k \rightarrow B} - T_{k \rightarrow B}^\dagger] - E_{K_B \rightarrow B} \\
&+ \alpha_1^{K_B \rightarrow B} - \alpha_3^{B \rightarrow P}, \\
T_{P' \rightarrow B'}^\dagger &= [T_{A \rightarrow P} - T_{Q \rightarrow P}^\dagger] + \alpha_1^{\{A,B,Q\} \rightarrow P} \\
&+ \alpha_4^P - \alpha_5^{B,P} - \alpha_6^{P \rightarrow B}.
\end{aligned} \tag{28}$$

B. The Zero-control Limit

The structural role of control variables α in the merge problem is recognized by analyzing the uncontrolled behavior of dynamical variables. In the zero-control limit with the blocking flow condition, all control variables vanish except those associated with blocked channels. Then, the equation (28) in the limit becomes,

$$T_{B' \rightarrow P'} = \sum_{k \in K_B \setminus \{P\}} [T_{k \rightarrow B} - T_{k \rightarrow B}^\dagger] - E_{K_B \rightarrow B}. \tag{29}$$

The transfer entropy $T_{B' \rightarrow P'}$ is determined solely by incoming information flows and is not constrained by the entropy $H(P)$. Since $T_{A' \rightarrow P'} = T_{A \rightarrow P} = \mu_A$ and $T_{Q' \rightarrow P'}^\dagger = T_{Q \rightarrow P}^\dagger = \mu_A$ by the zero-control limit,

$$\Delta H(P') = [T_{A' \rightarrow P'} + T_{B' \rightarrow P'} - T_{Q' \rightarrow P'}^\dagger] = T_{B' \rightarrow P'}. \tag{30}$$

Consequently, the formal zero-control evolution does not itself prevent continued increase of $H(P')$. However, such unrestricted growth is not generally realizable. For instance, if the component P has a binary state, then its entropy $H(P') \leq \ln 2$. In summary, the formal zero-control reduction contains no mechanism enforcing the finite-alphabet entropy bound, as the determination of transfer entropy $T_{B' \rightarrow P'}$ is independent of the entropy $H(P')$. Hence, the nonzero control variables α are required to introduce a finite maximum entropy of the component P , the merging point.

C. Near the Zero-control Limit with a fixed maximum entropy

A finite upper bound on the entropy $H(P') \leq \ln Q$ imposes constraints on the merging dynamics. Since $H(P'|P) \leq H(P') \leq \ln Q$, the regime $H(P'|P) \rightarrow \ln Q$ is examined, under the following assumptions:

$$\begin{aligned}
&\text{Assumption 1 : } T_{A' \rightarrow P'} - T_{\{A', B'\} \rightarrow P'} = I(A', P'' | P', B') = T_{A \rightarrow P} \\
&\text{Assumption 2 : } I(Q, P' | P, A) = 0, \quad I(Q', P'' | P', A', B') = 0 \\
&\text{Assumption 3 : } \alpha_1^{\{A, Q\} \rightarrow P} = \alpha_1^{\{A', B', Q'\} \rightarrow P'} \\
&\text{Assumption 4 : } \frac{\partial \alpha_1^{\{A, Q\} \rightarrow P}}{\partial \epsilon} > -1 \\
&\text{Assumption 5 : } Q' = P, Q'' = P' \\
&\text{Assumption 6 : } \Delta I(P', P) + T_{B' \rightarrow P'} \geq 0,
\end{aligned} \tag{31}$$

where ϵ is defined in Eq. (35).

These assumptions isolate a minimal class of interactions where merging behavior can be analytically characterized.

From Eq. (10),

$$\begin{aligned}
T_{B' \rightarrow P'} = & \Delta H(P') + T_{Q' \rightarrow P'}^\dagger - T_{A' \rightarrow P'} \\
& + E_{\{A', B', Q'\} \rightarrow P'} - \alpha_1^{\{A', B', Q'\} \rightarrow P'}.
\end{aligned} \tag{32}$$

The term $E_{\{A', B', Q'\} \rightarrow P'}$ can be reduced to the single multiple transfer entropy:

Proposition 4 *From Assumption 2, 5,*

$$E_{\{A', B', Q'\} \rightarrow P'} = T_{\{A', B'\} \rightarrow P'}. \tag{33}$$

(proof) Appendix C.

Proposition 4 and Assumption 1 lead to:

$$T_{B' \rightarrow P'} = \Delta H(P') + T_{Q' \rightarrow P'}^\dagger - T_{A \rightarrow P} - \alpha_1^{\{A', B', Q'\} \rightarrow P'}. \tag{34}$$

Here, we introduce $\epsilon \geq 0$ such that,

$$\epsilon := \ln Q - \alpha_1^{\{A, Q\} \rightarrow P} - T_{A \rightarrow P}. \tag{35}$$

The limit $\epsilon \rightarrow 0$ encodes that the sum of the transferred information from A and internal noise, $\alpha_1^{\{A,Q\} \rightarrow P}$, reaches the maximum $\ln Q$. Additionally, define δ by:

$$\delta := \ln Q - \Delta H(P') - T_{Q' \rightarrow P'}^\dagger, \quad (36)$$

indicating,

$$T_{B' \rightarrow P'} = \epsilon - \delta, \quad (37)$$

from Eq. (34) and Assumption 3.

Before discussing the relationship between ϵ and δ , one can deduce the lemma.

Lemma 1 *From Assumption 1, 2, 3, 5, and Proposition 4,*

$$H(P''|P') = H(P'|P) + T_{B' \rightarrow P'}. \quad (38)$$

(proof) Appendix D.

The Appendix D also establishes that Assumptions 2 and 5 imply $H(P'|P) = \alpha_1^{\{A,Q\} \rightarrow P} + T_{A \rightarrow P}$. Consequently, the limit $\epsilon \rightarrow 0$ corresponds to the saturation of the conditional-entropy ceiling: $\epsilon = \ln Q - H(P'|P)$.

Then,

Proposition 5 *From Assumption 5, 6 and Lemma 1,*

$$\begin{aligned} (a) \quad & H(P') \geq \ln Q - \epsilon \\ (b) \quad & 0 \leq \delta \leq \epsilon \\ (c) \quad & 0 \leq \Delta H(P') \leq \epsilon. \end{aligned} \quad (39)$$

(proof) Appendix E.

From Proposition 5, one obtains,

$$0 \leq T_{B' \rightarrow P'} \leq \epsilon = \ln Q - \alpha_1^{\{A,Q\} \rightarrow P} - T_{A \rightarrow P} = \ln Q - H(P'|P). \quad (40)$$

Equation (40) bounds the admissible external transfer entropy by ϵ , the remaining conditional-entropy margin. It therefore establishes competition at the level of an upper bound. Under Assumption 4, increasing the pre-existing transfer entropy reduces ϵ and hence lowers the admissible upper bound on the external transfer entropy. A corresponding decrease in the realized transfer entropy does not follow from the bound alone and must be verified for each specified model. The control variables α_3 and α_6 must accordingly satisfy the conditions leading to Eq. (40) as the conditional-entropy margin approaches zero.

In communication theory [45, 46], this competition is analogous to cross-talk in a two-sender-one-receiver multiple-access channel (MAC). The cross-talk [47–49] in MAC has been referred to as the difficulty of simultaneous communication with two senders, when the receiver’s capacity for information processing is limited. Hence, Eq. (40) states that, as the pre-existing transfer from A to P increases, the admissible upper bound on transfer from B decreases when $H(P'|P)$ is close to $\ln Q$. A multiplex receiver can avoid this competition by assigning separate state-space coordinates to the two inputs, as illustrated in Sec. V D 2.

Intuitively, a finite-capacity receiving component cannot accommodate arbitrarily many independent incoming influences. Eq. (40) makes this competition quantitative by bounding the additional transfer entropy in terms of the remaining conditional-entropy margin, expressing suppression of admissible external information transfer.

Furthermore, under limiting conditions, we derive a more tractable formulation, since the upper bound of Eq. (40) differs from the realized transfer entropy $T_{B' \rightarrow P'}$. The defining saturation condition is $H(P'|P) \rightarrow \ln Q$. In this limit, Proposition 5 additionally implies,

$$\begin{aligned}
 H(P') &\rightarrow \ln Q \\
 \Delta H(P') &\rightarrow 0 \\
 T_{Q' \rightarrow P'}^\dagger &\rightarrow \ln Q \\
 T_{B' \rightarrow P'} &\rightarrow 0 \\
 H(P''|P') &\rightarrow \ln Q.
 \end{aligned} \tag{41}$$

From this limit, $T_{B' \rightarrow P'}$ can be expanded as:

$$T_{B' \rightarrow P'}(\epsilon) = \frac{dT_{B' \rightarrow P'}}{d\epsilon} \Big|_{\epsilon=0^+} \epsilon + o(\epsilon), \tag{42}$$

assuming the differentiability of $T_{B' \rightarrow P'}$ near $\epsilon = 0$.

By Eq. (40),

$$\frac{dT_{B' \rightarrow P'}}{d\epsilon} \Big|_{\epsilon=0^+} \geq 0, \quad (43)$$

indicating that the derivative at $\epsilon = 0$ is non-negative. Moreover, Assumption 4 ensures from Eq. (35) that $dT_{A \rightarrow P}/d\epsilon < 0$; namely, increasing the pre-existing flow reduces the remaining entropy-capacity margin. Hence, under this one-sided boundary condition, the derivative ratio can be expressed as:

$$\frac{dT_{B' \rightarrow P'}}{dT_{A \rightarrow P}} \Big|_{\epsilon=0^+} = \frac{\frac{dT_{B' \rightarrow P'}}{d\epsilon} \Big|_{\epsilon=0^+}}{\frac{dT_{A \rightarrow P}}{d\epsilon} \Big|_{\epsilon=0^+}} \leq 0. \quad (44)$$

Up to this point, ϵ is defined by Eq. (35), while $T_{B' \rightarrow P'}$ has been expanded based on its limiting behavior. No finite-margin monotonicity follows from the boundary result alone. To examine finite-margin behavior, we consider a parameterized family $\{p_\theta\}_{\theta \in \Theta}$ of underlying probability distributions on a fixed alphabet. In this case, $T_{A \rightarrow P}$ and $T_{B' \rightarrow P'}$ can be denoted $T_{A \rightarrow P}(\theta)$ and $T_{B' \rightarrow P'}(\theta)$, respectively. Along such a parameterized family, one can expect that the following condition on their local relationship can be satisfied over a finite range of θ ,

$$\frac{dT_{B' \rightarrow P'}}{dT_{A \rightarrow P}}(\theta) = \frac{\frac{dT_{B' \rightarrow P'}(\theta)}{d\theta}}{\frac{dT_{A \rightarrow P}(\theta)}{d\theta}} \leq 0. \quad (45)$$

Eq. (45) characterizes local suppression of the newly incoming flow with $\frac{dT_{A \rightarrow P}}{d\theta} \neq 0$. In Section V D, we examine Eq. (45) for explicit statistical models.

D. Examples of merging information flows

The following examples examine separately whether the hypotheses of the boundary result (Eq. (31)) are satisfied and whether a particular model exhibits a negative relation, Eq. (45), over a finite parameter range.

1. Ternary Additive Receiver

Let the dynamics at time $t < t_1$ be,

$$\begin{aligned} A' &\sim p_a \\ B' &\sim p_b \\ P' &= A \\ Q' &= P, \end{aligned} \quad \text{for } t < t_1 \quad (46)$$

where $p_\theta := (1 - 2\theta, \theta, \theta)$ is a one-parameter distribution on $\mathbb{Z}_3$ with $0 < \theta < \frac{1}{3}$ and A' and B' belong to mutually independent i.i.d. time sequences distributed according to p_a and p_b , respectively.

At time $t \geq t_1$,

$$\begin{aligned} A'' &\sim p_a \\ B'' &\sim p_b \\ P'' &= A' + B' \pmod{3} \\ Q'' &= P', \end{aligned} \quad (47)$$

Then,

$$T_{A \rightarrow P} = H(A) = h(a), \quad (48)$$

where $h(x) := -(1 - 2x)\ln(1 - 2x) - (2x)\ln(x)$.

Therefore,

$$\frac{dT_{A \rightarrow P}}{da} = h'(a) = 2 \ln \frac{1 - 2a}{a} > 0. \quad (49)$$

Since $\mathbb{P}(P'' = 0) = 1 - 2(a + b - 3ab)$ and $\mathbb{P}(P'' = 1) = \mathbb{P}(P'' = 2) = a + b - 3ab$,

$$\begin{aligned} T_{B' \rightarrow P'} &= I(P'', B') = H(P'') - H(A') \\ &= h(a + b - 3ab) - h(a). \end{aligned} \quad (50)$$

This yields,

$$\frac{dT_{B' \rightarrow P'}}{da} = (1 - 3b)h'(a + b - 3ab) - h'(a) < 0 \quad (\because h''(x) < 0 \text{ for } 0 < x < \frac{1}{3} \text{ and } 0 < 1 - 3b < 1) \quad (51)$$

Hence,

$$\frac{dT_{B' \rightarrow P'}}{dT_{A \rightarrow P}} = \frac{\frac{dT_{B' \rightarrow P'}}{da}}{\frac{dT_{A \rightarrow P}}{da}} < 0. \quad (52)$$

For this dynamics, the equation (45) is valid for the overall range. Also, it can be shown that the assumptions of Eq. (31) are valid.

2. A Multiplex Receiver

On the other hand, most artificial networks, such as multi-source networks in communication theory, are designed to prevent the cross-talk effect for a reliable communication. This is accomplished by modeling the multiplex receiver to process input signals independently. Therefore, the multiplex receiver is assigned two independent state-space components, $P = (P_1, P_2)$. Then, the dynamics is set as,

$$\begin{aligned}
 A' &\sim \text{Uniform}(N_{Z_A}) \\
 B' &\sim \text{Uniform}(N_{Z_B}) \\
 P'_1 &= A \\
 P'_2 &= P_2 \\
 Q'_1 &= P_1 \\
 Q'_2 &= P_2,
 \end{aligned}
 \quad \text{for } t < t_1 \tag{53}$$

where $N_Z := \{0, 1, \dots, Z-1\}$ and $Z_A, Z_B \in \mathbb{N}$, and $\text{Uniform}(N_Z)$ is the uniform distribution for the set N_Z .

$$\begin{aligned}
 A'' &\sim \text{Uniform}(N_{Z_A}) \\
 B'' &\sim \text{Uniform}(N_{Z_B}) \\
 P''_1 &= A' \\
 P''_2 &= B' \\
 Q''_1 &= P'_1 \\
 Q''_2 &= P'_2.
 \end{aligned}
 \quad \text{for } t \geq t_1 \tag{54}$$

As a result,

$$\begin{aligned}
 T_{A \rightarrow P} &= \ln(Z_A) \\
 T_{B' \rightarrow P'} &= \ln(Z_B),
 \end{aligned} \tag{55}$$

and, thus, $T_{B' \rightarrow P'}$ is independent of Z_A and hence of $T_{A \rightarrow P}$ with a fixed Z_B . Since this construction avoids competition by supplying an additional receiver coordinate, it is a contrast model rather than a fixed-alphabet test of the boundary derivative. Consequently,

this represents a limiting case in which the cross-talk effect is eliminated.

3. AND Gate

Unlike the previous examples, the merge with AND gate violates some of the assumptions and does not satisfy Eq. (45) in general.

$$\begin{aligned}
 A' &\sim B\left(\frac{1}{2} + \rho_A\right) \\
 B' &\sim B\left(\frac{1}{2} + \rho_B\right) \quad \text{for } t < t_1 \\
 P' &= A \\
 Q' &= P,
 \end{aligned} \tag{56}$$

where $B(\rho)$ is the binomial distribution with probability ρ and $-\frac{1}{2} \leq \rho_A, \rho_B \leq \frac{1}{2}$.

$$\begin{aligned}
 A'' &\sim B\left(\frac{1}{2} + \rho_A\right) \\
 B'' &\sim B\left(\frac{1}{2} + \rho_B\right) \quad \text{for } t \geq t_1 \\
 P'' &= \text{AND}(A', B') \\
 Q'' &= P'.
 \end{aligned} \tag{57}$$

Here,

$$\begin{aligned}
 T_{A \rightarrow P} &= -\left(\frac{1}{2} + \rho_A\right) \ln\left(\frac{1}{2} + \rho_A\right) - \left(\frac{1}{2} - \rho_A\right) \ln\left(\frac{1}{2} - \rho_A\right) \\
 T_{B' \rightarrow P'} &= -\left(\frac{1}{2} + \rho_A\right)\left(\frac{1}{2} + \rho_B\right) \ln\left(\frac{1}{2} + \rho_B\right) + \left(\frac{1}{2} - \rho_A\right)\left(\frac{1}{2} + \rho_B\right) \ln\left(\frac{1}{2} - \rho_A\right) \\
 &\quad - \left(1 - \left(\frac{1}{2} + \rho_A\right)\left(\frac{1}{2} + \rho_B\right)\right) \ln\left(1 - \left(\frac{1}{2} + \rho_A\right)\left(\frac{1}{2} + \rho_B\right)\right).
 \end{aligned} \tag{58}$$

While,

$$\frac{\partial T_{B' \rightarrow P'}}{\partial \rho_A} = \left(\frac{1}{2} + \rho_B\right) \ln \frac{1 - \left(\frac{1}{2} + \rho_A\right)\left(\frac{1}{2} + \rho_B\right)}{\left(\frac{1}{2} - \rho_A\right)\left(\frac{1}{2} + \rho_B\right)} \geq 0, \tag{59}$$

$$\frac{\partial T_{A \rightarrow P}}{\partial \rho_A} = \ln \frac{\frac{1}{2} - \rho_A}{\frac{1}{2} + \rho_A} \begin{cases} < 0 & (\rho_A > 0) \\ > 0 & (\rho_A < 0) \\ = 0 & (\rho_A = 0) \end{cases} \tag{60}$$

Therefore,

$$\frac{dT_{B' \rightarrow P'}}{dT_{A \rightarrow P}} \begin{cases} \leq 0 & (\rho_A > 0) \\ \geq 0 & (\rho_A < 0) \end{cases} \quad (61)$$

Assumptions 2, 3, and 5 are satisfied; however, Assumption 1 fails except the trivial case where $\rho_B = \frac{1}{2}$, and the validity of Assumption 4 depends on the value of ρ_A and ρ_B .

VI. EXAMPLE

To illustrate the mechanism derived in Sec. V, we construct a stochastic network with the Fredkin gate [50], which is a standard gate in the literature of reversible computing [51–54]: a cycle $P \rightarrow A1 \rightarrow \dots \rightarrow A3 \rightarrow P$ and an external node (Ext). Each node has a two-bit state, $\Gamma_{A1}, \dots, \Gamma_{A3}, \Gamma_P, \Gamma_{Ext} \in \{(0, 0), (0, 1), (1, 0), (1, 1)\}$, and the update rule at the merging node P is given by a Fredkin-type controlled-SWAP gate:

$$\Gamma'_P = \begin{cases} (\gamma_{A3}^{(2)}, \gamma_{A3}^{(1)}) & \text{if } \gamma_{Ext}^{(1)} = 1 \\ (\gamma_{A3}^{(1)}, \gamma_{A3}^{(2)}) & \text{if } \gamma_{Ext}^{(1)} = 0 \end{cases}, \quad (62)$$

where $\Gamma_X = (\gamma_X^{(1)}, \gamma_X^{(2)})$ for a node X with $\gamma_X^{(1)}, \gamma_X^{(2)} \in \{0, 1\}$.

Although the gate originates from conservative logic, we analyze its information dynamics under a parameterized input ensemble. For $0 \leq \theta < 1/4$, we define the probability mass function p_θ such that, for a random variable $\Gamma_X \sim p_\theta$,

$$\begin{aligned} p_\theta(\Gamma_X = (1, 0)) &= \frac{1}{4} + \theta \\ p_\theta(\Gamma_X = (0, 1)) &= \frac{1}{4} - \theta \\ p_\theta(\Gamma_X = (0, 0)) &= p_\theta(\Gamma_X = (1, 1)) = \frac{1}{4}. \end{aligned} \quad (63)$$

Then, the initial ensemble at $t = t_0$ is constructed as:

$$\begin{aligned} \Gamma_{A1}, \Gamma_{A2}, \Gamma_{A3} &\sim p_\theta \\ \Gamma_P &\sim p_\theta \\ \Gamma_{Ext} &\sim p_0, \end{aligned} \quad (64)$$

where $\Gamma_{A1}, \dots, \Gamma_{A3}, \Gamma_P, \Gamma_{Ext}$ are mutually independent.

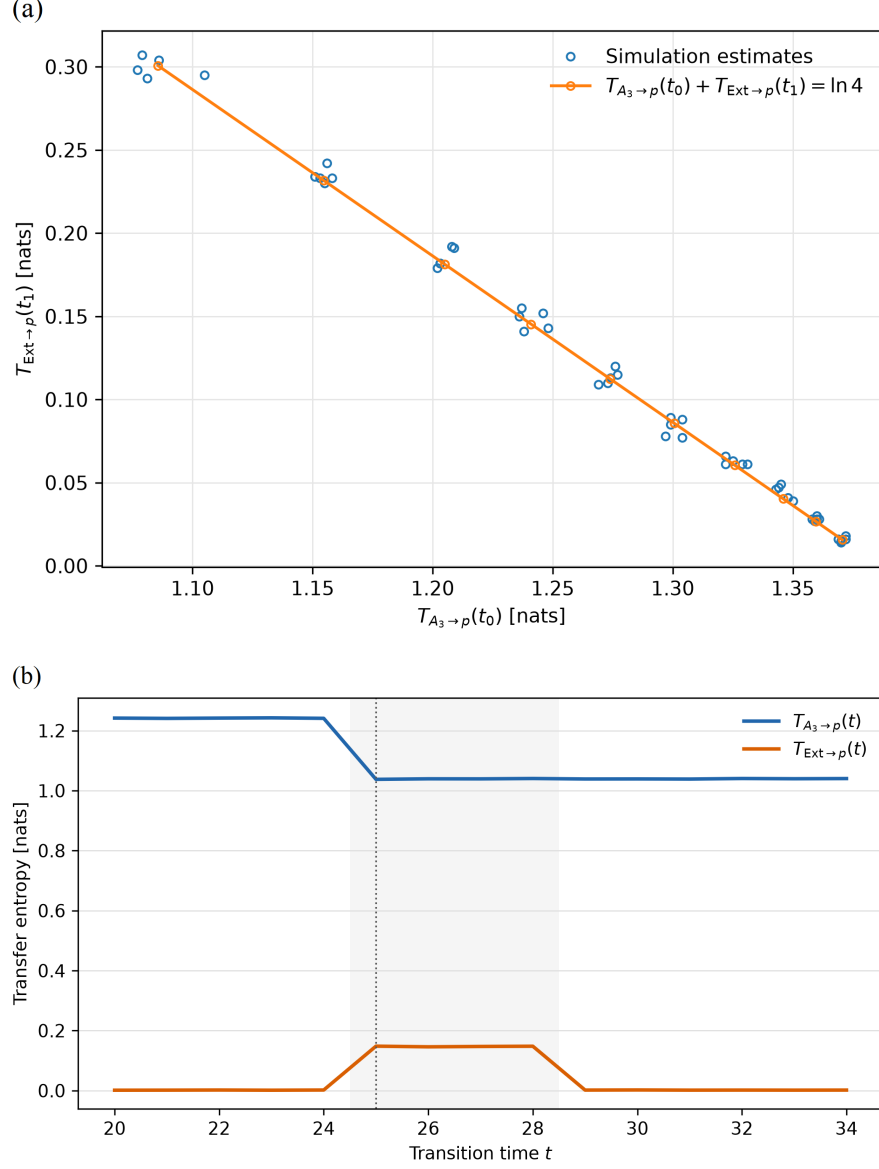


FIG. 3: **(a)** Simulation results for competing information flows: $T_{A_3 \rightarrow P}(t_0)$ versus $T_{Ext \rightarrow P}(t_1)$. The cycle $P \rightarrow A_1 \rightarrow \dots \rightarrow A_3 \rightarrow P$ is isolated at time $t = t_0$, and node P begins to interact with the external source (Ext) at $t = t_1$. **(b)** The time-evolution of $T_{A_3 \rightarrow P}$ and $T_{Ext \rightarrow P}$ with $\theta = 0.18$. Here, $t_1 = 25$ time steps.

The remaining nodes in the cycle are modeled as delay-copy units, such as $\Gamma'_{A_2} = \Gamma_{A_1}$ or $\Gamma'_{A_1} = \Gamma_P$, ensuring consistency with Assumption 5 in Sec. V. Within this construction, the cycle represents a pre-existing internal information circulation, while the node P acts as a finite-capacity processor where internal and external information flows compete.

A network of two-bit nodes is constructed with the cycle $P \rightarrow A1 \rightarrow \dots \rightarrow A3 \rightarrow P$ and the external source (Ext), and the component P and (Ext) begin to interact after time $t = t_1$. The resulting dynamics satisfies all assumptions of Eq. (31): for the construction in this section, Assumptions 1, 2, and 5 follow directly from the conditional bijectivity of the controlled-SWAP operation and the delay-copy relation. At the onset of interaction, independence of the node states gives $\alpha_1^{\{A,Q\} \rightarrow P} = \alpha_1^{\{A',B',Q'\} \rightarrow P'} = 0$, $\frac{\partial \alpha_1^{\{A,Q\} \rightarrow P}}{\partial \epsilon} = 0$, $\Delta I(P', P) = 0$, so Assumptions 3, 4, and 6 are also satisfied. Then, the exact calculation yields:

$$T_{A3 \rightarrow P}(t_0) + T_{Ext \rightarrow P}(t_1) = \ln 4, \quad (65)$$

where $T_{A3 \rightarrow P}(t_0) := T_{A3 \rightarrow P}$ and $T_{Ext \rightarrow P}(t_1) := T_{Ext' \rightarrow P'}$.

At the onset of interaction, Eq. (65) shows that the realized external transfer entropy equals the remaining capacity margin. Consequently, within this parameterized Fredkin model, increasing $T_{A3 \rightarrow P}(t_0)$ is accompanied by a corresponding decrease in $T_{Ext \rightarrow P}(t_1)$, thereby realizing the finite-range suppression relation in Eq. (45).

To elucidate the time-evolution of these quantities, we simulate an ensemble of independent realizations and estimate the relevant information-theoretic quantities using a simple binning method. Simulation results (Fig. 3 (a)) [44] show that increasing the internal circulating flow $T_{A3 \rightarrow P}(t_0)$ suppresses the external input $T_{Ext \rightarrow P}(t_1)$ in accordance with Eq. (65), quantitatively confirming the relation (Eq. (45)).

Furthermore, Fig. 3 (b) shows that both $T_{A3 \rightarrow P}(t)$ and $T_{Ext \rightarrow P}(t)$ evolve over time. Following activation of the interaction, $T_{Ext \rightarrow P}(t)$ initially increases to a plateau before dissipating to zero as the probability distribution mixed by (Ext) propagates through the cycle to node P . Since the swapping operation by (Ext) symmetrizes the asymmetry induced by parameter θ in the initial ensemble, the node P ceases to receive influence from node (Ext) under this symmetrized distribution. This demonstrates that the competition between information flows exhibits distinct temporal characteristics.

VII. CONCLUSION

This work establishes two information-dynamical results. First, the circulation of transfer entropy and reversed transfer entropy constitutes a necessary structural condition for stationary or periodic entropic dynamics in finite systems under the stated assumptions and fixed network support. Second, when a pre-existing circulating flow and an external flow merge at a finite-capacity node, the remaining conditional-entropy margin provides an upper bound on the admissible external transfer entropy. In this sense, the bound identifies a possible information-transfer constraint associated with cyclic organization. This mechanism corresponds to a cross-talk effect, in which limited entropy capacity constrains simultaneous information transfer.

For the Fredkin model considered here, the bound is attained at the onset of interaction, producing a model-specific negative relation between the circulating and external transfer entropies. The subsequent simulation shows that this relation is transient and need not remain saturated after the modified distribution propagates around the cycle. Consequently, the quantitative framework developed in this work facilitates an information-dynamical analysis of reversible logic gates.

The current analysis relies on several assumptions introduced in Sec. II and Eq. (31). While these assumptions facilitate tractable analysis of interacting information flows and yield baseline cases, their necessity and generality remain to be examined. Further work is required to relax these assumptions, explore a wider class of merging mechanisms, and investigate additional properties of information circulation, including its temporal structure and interactions between multiple cycles.

A further limitation concerns the admissibility of the information-dynamical evolution, as discussed in Section IV B and Ref. [26]. A complete characterization of the admissible region of the control variables, subject to Shannon inequalities, alphabet bounds, and temporal consistency of the underlying distributions, remains an open problem.

Furthermore, the present analysis assumes that the directed structure of transfer entropy

interactions remains sufficiently stable over the analyzed time interval. Extending the framework to adaptive networks, in which the interaction topology and effective information flow structure evolve dynamically, remains an important direction for future research.

Data Availability Statement

The data that support the findings of this study are openly available in Zenodo at <https://doi.org/10.5281/zenodo.22358615> (Version v0.1.4b).

Appendix A: A Counterexample to the Assumption of Section II D

Let the random variables u_1, u_2, v represent the state of three components in discrete time having the dynamics:

$$u'_1, u'_2 \sim B\left(\frac{1}{2}\right) \quad v' = u_1 \oplus u_2, \quad (\text{A1})$$

where $B(p)$ is the binomial distribution with probability p , and the operation $\oplus$ represents the binary operation XOR.

Then, $H(v'|v) = H(v'|v, u_1) = H(v'|v, u_2) = \ln 2$, resulting in $T_{u_1 \rightarrow v} = T_{u_2 \rightarrow v} = 0$. Nonetheless, for $\mathbf{u} = (u_1, u_2)$, $T_{\mathbf{u} \rightarrow v} = H(v'|v) - H(v'|v, u_1, u_2) = \ln 2$.

Appendix B: The Proof of Proposition 3

From the intrinsic method of Ref. [26], $\alpha_{3456}^{\Gamma_j \rightarrow \Gamma_i} := -\alpha_3^{\Gamma_j \rightarrow \Gamma_i} - \alpha_4^{\Gamma_j} + \alpha_5^{\Gamma_i, \Gamma_j} + \alpha_6^{\Gamma_j \rightarrow \Gamma_i}$ can be expressed as:

$$\mathbf{x} = \mathbf{A}^{-1} \mathbf{y} - \mathbf{A}^{-1} \mathbf{b}, \quad (\text{B1})$$

where $\mathbf{y}, \mathbf{x}, \mathbf{b} \in \mathbb{R}^n$ and $\mathbf{A} \in \mathbb{R}^{n \times n}$ such that,

$$\begin{aligned} (\mathbf{y})_l &= \alpha_3^{\Gamma_i \rightarrow \Gamma_{j_l}} - \alpha_6^{\Gamma_i \rightarrow \Gamma_{j_l}} - \Delta \alpha_2^{\Gamma_i, \Gamma_{j_l}} \\ (\mathbf{x})_l &= \alpha_{3456}^{\Gamma_{j_l} \rightarrow \Gamma_i} \\ (\mathbf{b})_l &= \Delta E_{K_i \rightarrow \Gamma_i} - \Delta \alpha_1^{K_i \rightarrow \Gamma_i} \\ (\mathbf{A})_{lm} &= -1 + \delta_{lm}, \end{aligned} \quad (\text{B2})$$

Here, the component i 's neighbor set is denoted as $K_i = \{\Gamma_{j_1}, \dots, \Gamma_{j_n}\}$ with $n > 1$.

Therefore,

($i \neq p$ case)

$$\begin{aligned}\alpha_{3456}^{\Gamma_{i-1} \rightarrow \Gamma_i} &= -\alpha_3^{\Gamma_i \rightarrow \Gamma_{i+1}} + \alpha_6^{\Gamma_i \rightarrow \Gamma_{i+1}} + \Delta E^{\{\Gamma_{i-1}, \Gamma_{i+1}\} \rightarrow \Gamma_i} \\ &\quad - \Delta\alpha_1^{\Gamma_i} + \Delta\alpha_2^{\Gamma_i, \Gamma_{i+1}} \\ \alpha_{3456}^{\Gamma_{i+1} \rightarrow \Gamma_i} &= -\alpha_3^{\Gamma_i \rightarrow \Gamma_{i-1}} + \alpha_6^{\Gamma_i \rightarrow \Gamma_{i-1}} + \Delta E^{\{\Gamma_{i-1}, \Gamma_{i+1}\} \rightarrow \Gamma_i} \\ &\quad - \Delta\alpha_1^{\Gamma_i} + \Delta\alpha_2^{\Gamma_i, \Gamma_{i-1}}.\end{aligned}\tag{B3}$$

($i = p$ case)

$$\begin{pmatrix} \alpha_{3456}^{\Gamma_{p-1} \rightarrow \Gamma_p} \\ \alpha_{3456}^{\Gamma_{p+1} \rightarrow \Gamma_p} \\ \alpha_{3456}^{\Gamma_{Ext} \rightarrow \Gamma_p} \end{pmatrix} = \frac{1}{2} \begin{pmatrix} +1 & -1 & -1 \\ -1 & +1 & -1 \\ -1 & -1 & +1 \end{pmatrix} \mathbf{y} + \frac{1}{2} \mathbf{b},\tag{B4}$$

where,

$$\begin{aligned}\mathbf{y} &= \begin{pmatrix} \alpha_3^{\Gamma_p \rightarrow \Gamma_{p-1}} - \alpha_6^{\Gamma_p \rightarrow \Gamma_{p-1}} - \Delta\alpha_2^{\Gamma_p, \Gamma_{p-1}} \\ \alpha_3^{\Gamma_p \rightarrow \Gamma_{p+1}} - \alpha_6^{\Gamma_p \rightarrow \Gamma_{p+1}} - \Delta\alpha_2^{\Gamma_p, \Gamma_{p+1}} \\ \alpha_3^{\Gamma_p \rightarrow \Gamma_{Ext}} - \alpha_6^{\Gamma_p \rightarrow \Gamma_{Ext}} - \Delta\alpha_2^{\Gamma_p, \Gamma_{Ext}} \end{pmatrix}, \\ \mathbf{b} &= [\Delta E_{\{\Gamma_{p-1}, \Gamma_{p+1}, \Gamma_{Ext}\} \rightarrow \Gamma_p} - \Delta\alpha_1^{\{\Gamma_{p-1}, \Gamma_{p+1}, \Gamma_{Ext}\} \rightarrow \Gamma_p}] \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.\end{aligned}\tag{B5}$$

Substituting these relations into the dynamical equations (Eq. (10)) and imposing the blocking flows condition, $T_{\Gamma_i \rightarrow \Gamma_{i-1}}(t) = 0$, $T_{\Gamma_i \rightarrow \Gamma_{i+1}}^\dagger(t) = 0$, $T_{\Gamma_p \rightarrow \Gamma_{Ext}}(t) = 0$, $T_{\Gamma_{Ext} \rightarrow \Gamma_p}^\dagger(t) = 0$ for $\forall t, i$, yields,

$$\begin{aligned}
\Delta T_{\Gamma_i \rightarrow \Gamma_{i+1}} &= T_{\Gamma_{i-1} \rightarrow \Gamma_i} - T_{\Gamma_i \rightarrow \Gamma_{i+1}} - E_{\{\Gamma_{i-1}, \Gamma_{i+1}\} \rightarrow \Gamma_i} + \alpha_1^{\{\Gamma_{i-1}, \Gamma_{i+1}\} \rightarrow \Gamma_i} \\
&\quad - \alpha_2^{\Gamma_i, \Gamma_{i+1}} - \alpha_3^{\Gamma_i \rightarrow \Gamma_{i+1}} \quad (i \neq p) \\
\Delta T_{\Gamma_i \rightarrow \Gamma_{i+1}}^\dagger &= 0 \\
\Delta T_{\Gamma_i \rightarrow \Gamma_{i-1}} &= 0 \\
\Delta T_{\Gamma_i \rightarrow \Gamma_{i-1}}^\dagger &= T_{\Gamma_{i-1} \rightarrow \Gamma_{i-2}}^\dagger - T_{\Gamma_i \rightarrow \Gamma_{i-1}}^\dagger - E'_{\{\Gamma_{i-2}, \Gamma_i\} \rightarrow \Gamma_{i-1}} + \alpha_1^{\{\Gamma_{i-2}, \Gamma_i\} \rightarrow \Gamma_{i-1}} \\
&\quad - \alpha_2^{\Gamma_{i-1}, \Gamma_i} - \alpha_6^{\Gamma_{i-1} \rightarrow \Gamma_{i-2}} \quad (i \neq p+1) \\
\Delta T_{\Gamma_p \rightarrow \Gamma_{p+1}} &= T_{\Gamma_{p-1} \rightarrow \Gamma_p} + T_{\Gamma_{Ext} \rightarrow \Gamma_p} - T_{\Gamma_p \rightarrow \Gamma_{p+1}} - E_{\{\Gamma_{p-1}, \Gamma_{p+1}, \Gamma_{Ext}\} \rightarrow \Gamma_p} \\
&\quad + \alpha_1^{\{\Gamma_{p-1}, \Gamma_{p+1}, \Gamma_{Ext}\} \rightarrow \Gamma_p} - \alpha_2^{\Gamma_p, \Gamma_{p+1}} - \alpha_3^{\Gamma_p \rightarrow \Gamma_{p+1}} \\
\Delta T_{\Gamma_{p+1} \rightarrow \Gamma_p}^\dagger &= \alpha_1^{\{\Gamma_p\} \rightarrow \Gamma_{Ext}} - \alpha_3^{\Gamma_{Ext} \rightarrow \Gamma_p} + T_{\Gamma_p \rightarrow \Gamma_{p-1}}^\dagger - T_{\Gamma_{p+1} \rightarrow \Gamma_p}^\dagger - \alpha_6^{\Gamma_p \rightarrow \Gamma_{p-1}} \\
&\quad - E'_{\{\Gamma_{p-1}, \Gamma_{p+1}, \Gamma_{Ext}\} \rightarrow \Gamma_p} + \alpha_1^{\{\Gamma_{p-1}, \Gamma_{p+1}, \Gamma_{Ext}\} \rightarrow \Gamma_p} - \alpha_2^{\Gamma_p, \Gamma_{p-1}} \\
\Delta T_{\Gamma_{Ext} \rightarrow \Gamma_p} &= \alpha_1^{\{\Gamma_p\} \rightarrow \Gamma_{Ext}} - T_{\Gamma_{Ext} \rightarrow \Gamma_p} - \alpha_3^{\Gamma_{Ext} \rightarrow \Gamma_p} \\
\Delta T_{\Gamma_p \rightarrow \Gamma_{Ext}}^\dagger &= -\alpha_{3456}^{\Gamma_p \rightarrow \Gamma_{Ext}},
\end{aligned} \tag{B6}$$

where E' denotes E evaluated at the following time step.

As Ref. [26], it is required to analyze near the zero-control limit to examine the effect of merging. Therefore,

$$\begin{aligned}
\alpha_1^{\{\Gamma_{i-1}, \Gamma_{i+1}\} \rightarrow \Gamma_i} &= \alpha_2^{\Gamma_i, \Gamma_{i+1}} = 0 \quad (i \neq p) \\
\alpha_1^{\{\Gamma_{i-1}, \Gamma_{i+1}\} \rightarrow \Gamma_i} &= \alpha_2^{\Gamma_i, \Gamma_{i+1}} = 0 \quad (i \neq p) \\
\alpha_1^{\{\Gamma_{p-1}, \Gamma_{p+1}, \Gamma_{Ext}\} \rightarrow \Gamma_p} &= \alpha_1^{\{\Gamma_{p-1}, \Gamma_{p+1}, \Gamma_{Ext}\} \rightarrow \Gamma_p} = \alpha_2^{\Gamma_i, \Gamma_{i+1}} = \alpha_2^{\Gamma_i, \Gamma_{i+1}} = 0 \\
\alpha_3^{\Gamma_i \rightarrow \Gamma_{i+1}} &= 0 \quad (i \neq p-1) \\
\alpha_6^{\Gamma_{i-1} \rightarrow \Gamma_{i-2}} &= 0 \quad (i \neq p+1) \\
\alpha_3^{\{\Gamma_{i-1}, \Gamma_{i+1}\} \rightarrow \Gamma_i} &= \alpha_5^{\{\Gamma_{i-1}, \Gamma_{i+1}\}} = \alpha_5^{\{\Gamma_{i-1}, \Gamma_i, \Gamma_{i+1}\}} = \alpha_6^{\{\Gamma_{i-1}, \Gamma_{i+1}\} \rightarrow \Gamma_i} = 0 \quad (i \neq p),
\end{aligned} \tag{B7}$$

retaining only the control variables associated with the newly opened channel between Ext and p . In this limit,

$$\Delta E = -\alpha_3^{\{\Gamma_{i-1}, \Gamma_{i+1}\} \rightarrow \Gamma_i} - \alpha_5^{\{\Gamma_{i-1}, \Gamma_{i+1}\}} + \alpha_5^{\{\Gamma_{i-1}, \Gamma_i, \Gamma_{i+1}\}} + \alpha_6^{\{\Gamma_{i-1}, \Gamma_{i+1}\} \rightarrow \Gamma_i} = 0, \tag{B8}$$

as the high-order control variables also vanish. Together with the stated initial condition, this implies $E_{\{\Gamma_{i-1}, \Gamma_{i+1}\} \rightarrow \Gamma_i} = E'_{\{\Gamma_{i-1}, \Gamma_{i+1}\} \rightarrow \Gamma_i} = 0$ for $i \neq p$ throughout the interval considered.

This yields Eq. (25).

■

The equation (25) requires further specification. Firstly, $\alpha_{3456}^{\Gamma_p \rightarrow \Gamma_{Ext}}$ cannot be replaced by α_3 , α_6 by the intrinsic method introduced in Ref. [26], since the component Ext has only one neighbor. Therefore, $\alpha_{3456}^{\Gamma_p \rightarrow \Gamma_{Ext}}$ has to be determined from its definition, $\alpha_{3456}^{\Gamma_p \rightarrow \Gamma_{Ext}} = -\alpha_3^{\Gamma_p \rightarrow \Gamma_{Ext}} - \alpha_4^{\Gamma_p} + \alpha_5^{\Gamma_p, \Gamma_{Ext}} + \alpha_6^{\Gamma_p \rightarrow \Gamma_{Ext}}$. Secondly, defining the dynamic characteristics of the component Ext can specify the terms, $\alpha_5^{\Gamma_p, \Gamma_{Ext}}$ and $\alpha_1^{\{\Gamma_p\} \rightarrow \Gamma_{Ext}}$. For instance, $\alpha_5^{\Gamma_p, \Gamma_{Ext}} = 0$ at all times and $\alpha_1^{\{\Gamma_p\} \rightarrow \Gamma_{Ext}} = 0$ at $t < t_1$ if the component Ext is regarded as a source of white noise.

On the other hand, if $E_{\{\Gamma_{p-1}, \Gamma_{p+1}, \Gamma_{Ext}\} \rightarrow \Gamma_p} = 0$ permanently, there is no stationary state with nonzero $T_{\Gamma_{Ext} \rightarrow \Gamma_p}$, $T_{\Gamma_p \rightarrow \Gamma_{Ext}}^\dagger$. This shows that multivariate effects are essential for the existence of a stationary state with nonzero interaction with the external source.

Such transient effects are expected to appear in explicit dynamical realizations of the merging process. The transient behaviors after time $t = t_1$ and the way to approach another stationary state were demonstrated numerically in Ref. [44].

Appendix C: The Proof of Proposition 4

By the definition of E ,

$$\begin{aligned} E_{\{A', B', Q'\} \rightarrow P'} &= [T_{\{A', B'\} \rightarrow \{P'\}} - T_{\{A', B'\} \rightarrow \{P'\}}^\dagger] + [T_{\{A', Q'\} \rightarrow \{P'\}} - T_{\{A', Q'\} \rightarrow \{P'\}}^\dagger] \\ &\quad + [T_{\{B', Q'\} \rightarrow \{P'\}} - T_{\{B', Q'\} \rightarrow \{P'\}}^\dagger] - [T_{\{A', B', Q'\} \rightarrow \{P'\}} - T_{\{A', B', Q'\} \rightarrow \{P'\}}^\dagger]. \end{aligned} \quad (C1)$$

From Assumption 5, one obtains,

$$\begin{aligned} T_{\{A', Q'\} \rightarrow \{P'\}}^\dagger &= I(\{A'', Q'', P'\} | P'') = I(A'', Q'' | P'') - I(A'', Q'' | P' P'') \\ &= I(A'', P' | P'') \quad (\because \text{Assumption 5: } Q'' = P') \\ &= T_{A' \rightarrow P'}^\dagger = 0 \quad (\because \text{blocking flows condition}), \end{aligned} \quad (C2)$$

and $T_{\{B', Q'\} \rightarrow \{P'\}}^\dagger = 0$ similarly. In addition,

$$\begin{aligned}
T_{\{A', B', Q'\} \rightarrow \{P'\}}^\dagger &= I(\{A'', B'', Q'', P'\} | P'') = I(\{A'', B'', P'\} | P'') - I(\{A'', B'', P'\} | P'' Q'') \\
&= I(\{A'', B'', P'\} | P'') \quad (\because \text{Assumption 5: } Q'' = P') \\
&= T_{\{A', B'\} \rightarrow \{P'\}}^\dagger.
\end{aligned} \tag{C3}$$

The $T_{\{A', B', Q'\} \rightarrow \{P'\}}$ is evaluated using Assumption 2:

$$\begin{aligned}
T_{\{A', B', Q'\} \rightarrow \{P'\}} &= I(\{A', B', Q', P''\} | P') = I(\{A', Q', P''\} | P') - I(\{A', Q', P''\} | P' B') \\
&= T_{\{A', Q'\} \rightarrow \{P'\}} - I(Q', P'' | P' B') + I(Q', P'' | P' A' B') \\
&= T_{\{A', Q'\} \rightarrow \{P'\}} + [I(Q', P'' | P') - I(Q', P'' | P' B')] + I(Q', P'' | P' A' B') \\
&\quad (\because I(Q', P'' | P') = T_{Q' \rightarrow P'} = 0 \text{ blocking flows condition}) \\
&= T_{\{A', Q'\} \rightarrow \{P'\}} + T_{\{B', Q'\} \rightarrow \{P'\}} \\
&\quad (\because \text{Assumption 2: } I(Q', P'' | P' A' B') = H(P'' | P' A' B') - H(P'' | P' A' B' Q') = 0).
\end{aligned} \tag{C4}$$

Combining the preceding results, the Eq. (C1) becomes,

$$E_{\{A', B', Q'\} \rightarrow P'} = T_{\{A', B'\} \rightarrow \{P'\}}. \tag{C5}$$

■

Appendix D: The Proof of Lemma 1

Under Assumptions 2 and 5,

$$\begin{aligned}
\alpha_1^{\{A, Q\} \rightarrow P} &= H(P' | P A Q) - H(P | P' A' Q') = H(P' | P A) \\
\alpha_1^{\{A', B', Q'\} \rightarrow P'} &= H(P'' | P' A' B' Q') - H(P' | P'' A'' B'' Q'') = H(P'' | P' A' B').
\end{aligned} \tag{D1}$$

Since,

$$\begin{aligned}
H(P' | P) &= T_{A \rightarrow P} + H(P' | P A) \\
&= T_{A \rightarrow P} + \alpha_1^{\{A, Q\} \rightarrow P},
\end{aligned} \tag{D2}$$

$$\begin{aligned}
H(P''|P') &= T_{B' \rightarrow P'} + T_{A' \rightarrow P'} - T_{\{A', B'\} \rightarrow \{P'\}} + H(P''|P' A' B') \\
&= T_{B' \rightarrow P'} + T_{A \rightarrow P} + H(P''|P' A' B') (\because \text{Assumption 1}) \\
&= T_{B' \rightarrow P'} + T_{A \rightarrow P} + H(P'|PA) (\because \text{Assumption 3}) \\
&= T_{B' \rightarrow P'} + H(P'|P).
\end{aligned} \tag{D3}$$

■

Appendix E: The Proof of Proposition 5

(a)

From Eq. (D2),

$$\begin{aligned}
H(P'|P) &= T_{A \rightarrow P} + \alpha_1^{\{A, Q\} \rightarrow P} \\
&= \ln Q - \epsilon,
\end{aligned} \tag{E1}$$

one obtains,

$$H(P') \geq H(P'|P) = \ln Q - \epsilon. \tag{E2}$$

(b)

From Assumption 5,

$$T_{Q' \rightarrow P'}^\dagger = H(Q''|P'') - H(Q''|P' P'') = H(P'|P'') = -\Delta H(P') + H(P''|P'). \tag{E3}$$

As a result,

$$H(P''|P') = \ln Q - \delta. \tag{E4}$$

Since $H(P''|P') \leq H(P'') \leq \ln Q$, one obtains $0 \leq \delta$.

Here, $\delta \leq \epsilon$ is derived from the non-negativity of $T_{B' \rightarrow P'}$.

(c)

From Assumption 6 and Lemma 1,

$$\Delta I(P', P) = \Delta H(P') - \Delta H(P'|P) = \Delta H(P') - T_{B' \rightarrow P'} \geq -T_{B' \rightarrow P'}. \tag{E5}$$

Therefore,

$$\Delta H(P') \geq 0. \tag{E6}$$

Then,

$$\ln Q \geq H(P'') \geq H(P') \geq \ln Q - \epsilon. \quad (\text{E7})$$

Hence,

$$\Delta H(P') \leq \ln Q - (\ln Q - \epsilon) = \epsilon. \quad (\text{E8})$$

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